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Cryptolets: An Open-Source Hardware Framework for Cryptographic Kernels

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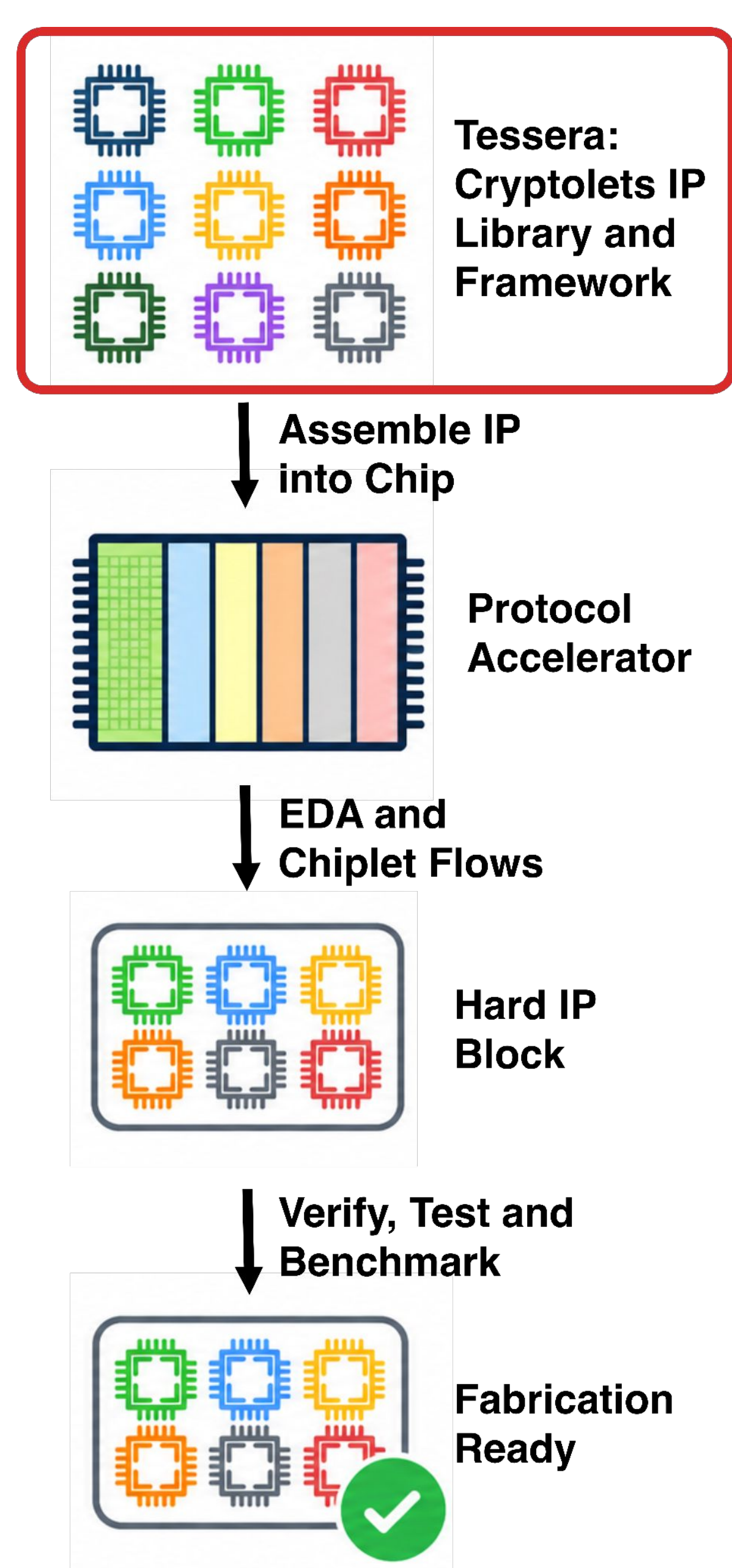
What is Cryptolets?

Cryptographic computing promises applications such as private machine-learning inference and verifiable outsourced computation, but its substantial computational overhead limits practical adoption. Prior work developed dedicated chips to address this overhead, but research groups often operate in silos. The expertise and effort required to build, verify, and evaluate these systems create high barriers to entry.

Cryptolets is an open-source research initiative addressing these challenges through a shared library of arithmetic primitives, cryptographic kernels, and accelerator prototypes, supported by automated design workflows, integration tools, and verification infrastructure. By providing reusable hardware and a common foundation for experimentation, it aims to lower barriers to entry, improve reproducibility, and help researchers build on one another's work. The initiative is led by NYU in collaboration with Stanford and CUNY, with support from NSF.

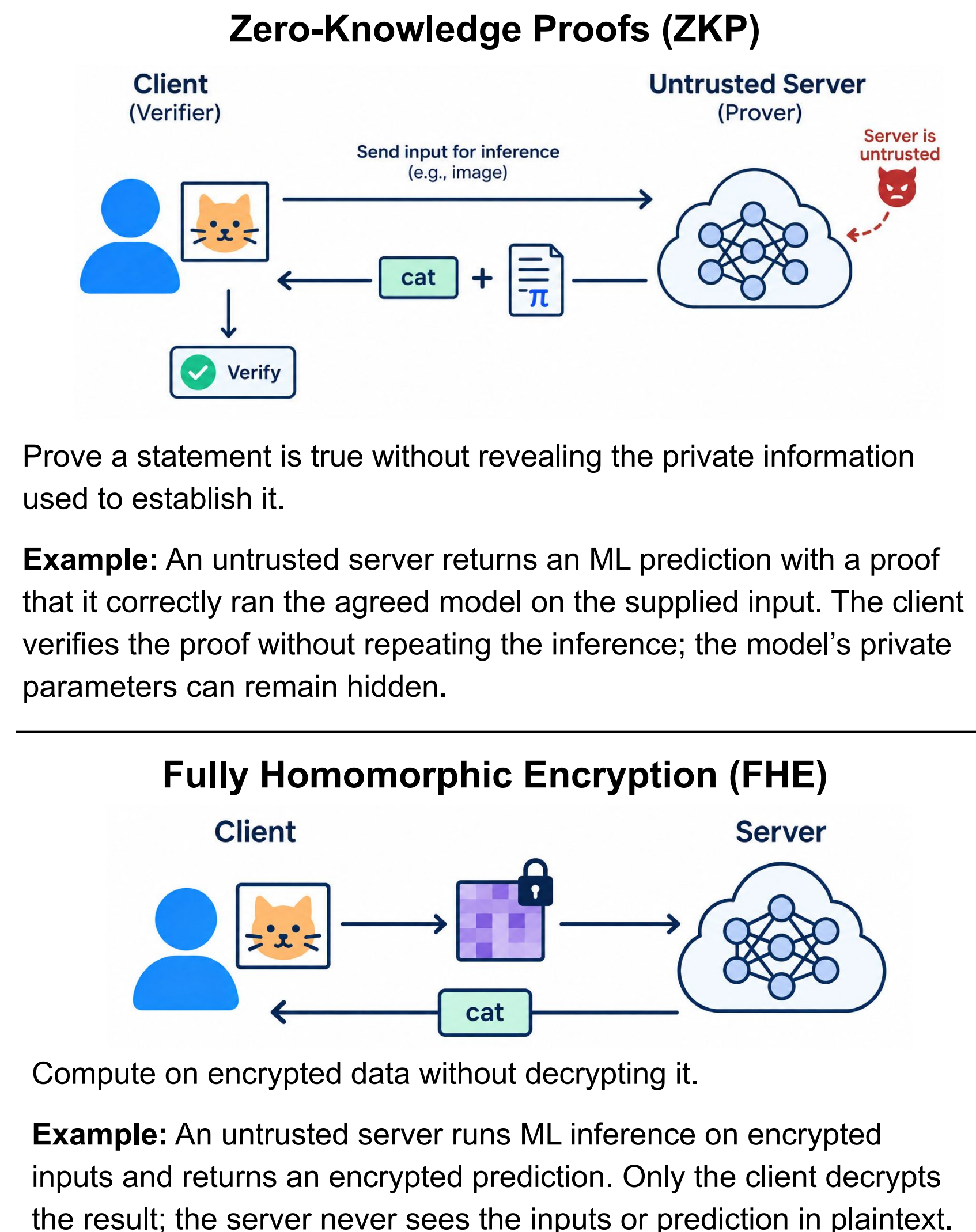
Tessera

Tessera is the open-source framework and hardware library within Cryptolets for developing, exploring, and sharing cryptographic hardware. It combines reusable, parameterized HLS C++ implementations with an automated end-to-end flow for hardware generation, verification, synthesis, and power analysis across ASIC and FPGA targets. The framework enables large-scale design-space exploration, while hierarchical composition and reusable design packages allow researchers to build larger kernels from existing hardware. Together, the library and framework lower barriers to entry and provide a shared foundation for reproducible, collaborative research.



Cryptographic Computing

Cryptographic computing enables computation on sensitive data while preserving privacy and verifiability. Two key approaches are zero-knowledge proofs (ZKP) and fully homomorphic encryption (FHE).



Cryptolets IP Library

Class	Level	IP	Done	Variants	ZKP	FHE	cMPC
Primitives	0	IntMul	✓	Schoolbook	×	×	×
		IntSq	✓	Karatsuba	×	×	×
		Const Mul	✓		×	×	×
		ModMul	✓	Montgomery	×	×	×
		ModInv	✓	Barrett	×	×	×
	1	ModAdd	✓		×	×	×
		ModSub	✓		×	×	×
		Const ModMul	✓		×	×	×
		ModInv	✓		×	×	×
		PADD	✓	Short Weierstrass	×		
Kernel	2	PDBL	✓	Twisted Edwards	×		
		AES	✓	Short Weierstrass	×		
		SHA-3			×		×
		FXOR				×	×
		Half-Gate					×
	2	NTT	✓	Cooley-Tukey	×	×	
				Pease			
				Korn-Lambiotte			
				Stockham			
		BConv				×	
		Auto.				×	
		PRNG				×	
		Had.			×	×	×
		MSM			×		
		Sumcheck			×		
		MLE			×		
		MTU			×		
		OT					×

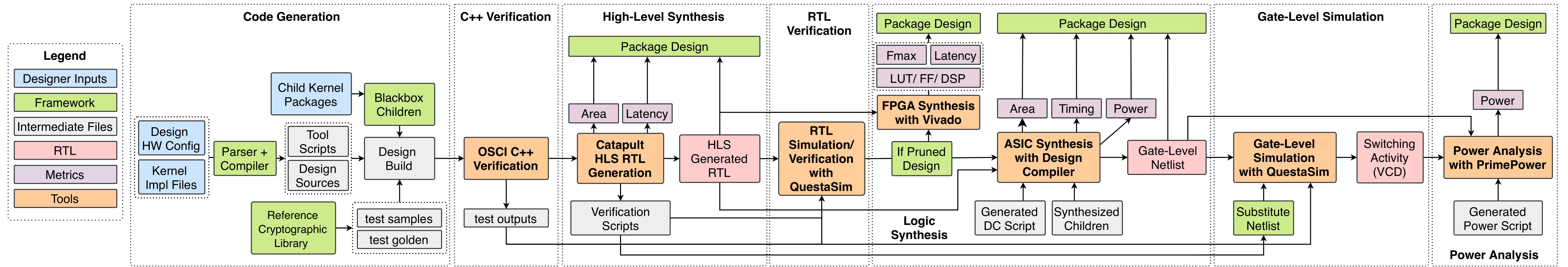
4901 Lines of
Source Code

13 IP
Modules
Implemented

+195 Million
Lines of RTL
Generated

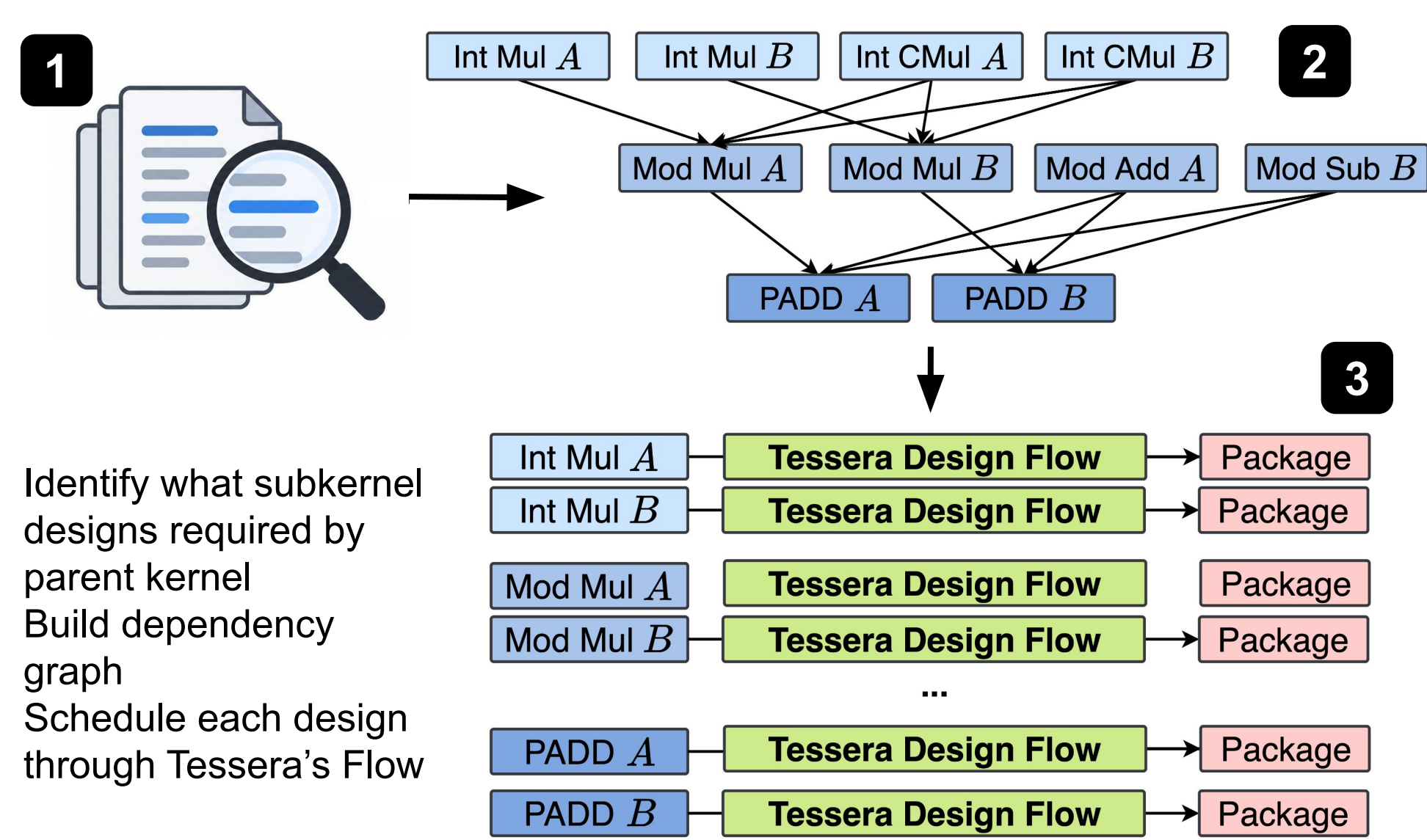
The Cryptolets IP Library supports FHE, ZKP, and cMPC (cryptographic multiparty computation), progressing from integer arithmetic (Level 0) to modular arithmetic (Level 1), then elliptic-curve operations and NTTs (Level 2). The table summarizes variants, application relevance, and implementation status, including ongoing and planned development.

Tessera End-to-End Design Flow

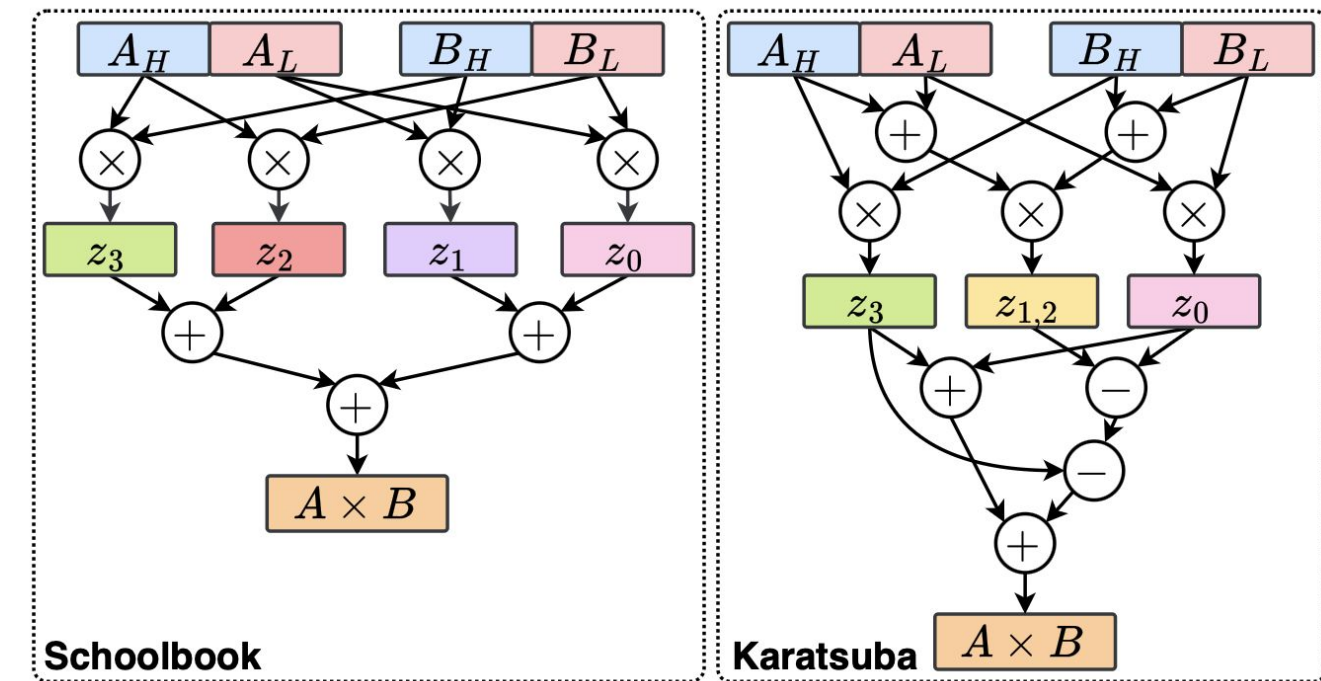


Tessera automates hardware generation, verification, synthesis, and power analysis. First, design sources and scripts are generated, incorporating child kernels as blackboxes. The HLS C++ implementation is verified against a cryptographic reference library before HLS generates RTL, which is verified through simulation. For ASICs, logic synthesis produces a netlist for gate-level verification and activity-based power analysis. For FPGAs, Vivado synthesis provides resource and timing estimates. Design configurations, RTL, and metrics are packaged for sharing and reuse.

Hierarchical Kernel Composition



Multiplier Optimizations



Large integer multiplications (254–754 bits) are a cryptographic hardware bottleneck. Tessera implements recursive Schoolbook and Karatsuba decomposition in HLS; Karatsuba trades extra additions for fewer multiplications, exposing area–latency tradeoffs. These building blocks form **modular multipliers** using **Montgomery** or **Barrett** reduction.

Methodology

Hardware Implementation. We characterize a Catapult HLS library with GlobalFoundries 12 nm standard cells using Synopsys Design Compiler Y-2026.03. Siemens Catapult HLS Ultra 2026.1 generates RTL from parameterized C++ kernels. Design Compiler estimates ASIC area and timing; PrimePower Y-2026.03 estimates power from gate-level switching activity. FPGA synthesis uses AMD Vivado 2026.1 through Catapult.

Testing and Verification. Python reference implementations generate test inputs and expected outputs for integer, modular, and elliptic-curve arithmetic. A shared C++ testbench checks C++ behavior through Catapult's OSCI flow and RTL and gate-level behavior through Siemens QuestaSim 2026.2, with configurable test counts per sweep.

Evaluation: Point Addition (Locus, ICCAD 2026)

Tessera began as **Locus**, which optimizes elliptic-curve point addition (PADD), a core operation in Multi-Scalar Multiplication (MSM) and a major ZKP proving bottleneck. Locus explores over 1,000 PADD designs and evaluates their impact on MSM and proof generation.

Figure 4 our framework enables large-scale design-space exploration across elliptic curves, identifying Pareto-optimal designs for different area and latency requirements.

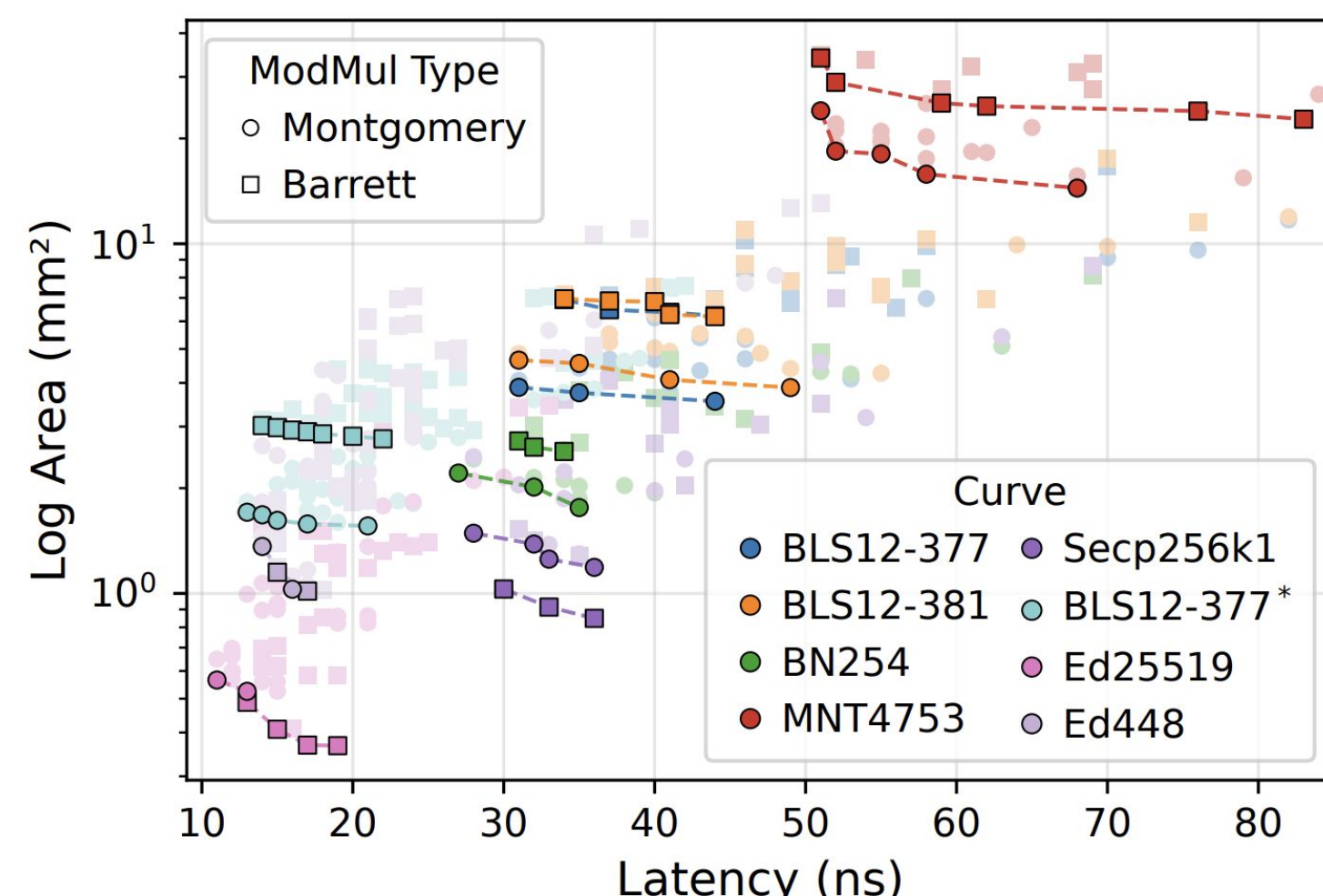


Figure 4: Design Space of PADD across named curves with Pareto-optimal points highlighted.

Table 2: Design Knobs for Best Performance Designs

Curve	Montgomery						Barrett					
	ct	qt	ret	kar	ct	qt	ct	qt	ret	kar	ct	qt
BN254	-	F	V	0	-	V	F	2	-	V	F	0
BLS12-377	-	F	F	2	-	V	F	3	-	F	F	2
BLS12-381	-	F	F	2	-	V	F	3	-	F	F	2
MNT4753	-	V	F	0	-	V	F	4	-	F	F	3
Secp256k1	-	F	V	0	-	F	F	2	-	F	F	0
Ed25519	F	F	F	1	F	F	F	2	F	F	F	0
BLS12-377*	F	F	F	2	V	V	F	3	F	F	F	2
Ed448	F	F	F	1	F	F	F	3	F	F	F	2

Table 2 highlights the configurations for the fastest and smallest PADDs for each curve. Optimal settings depend on the curve, and are non-obvious, motivating the need for a framework.

Figure 5 shows an ablation study for the fastest and smallest PADD designs.

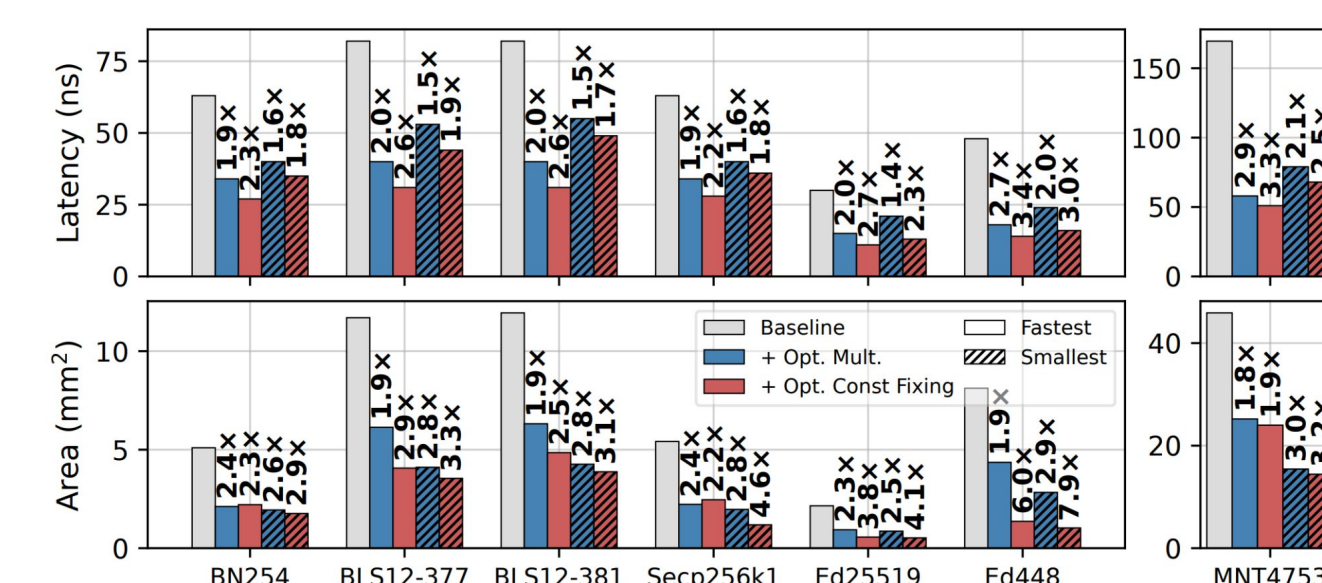


Figure 5: Ablation Study: latency and area for each optimization applied cumulatively over the baseline, for fastest and smallest Locus PADDs. Annotations show speedups and area reductions with respect to the baseline.

On **ASICs**, we framework achieves **2.71× geomean speedup** for its fastest designs and **3.11× geomean area reduction** for its smallest over reproduced prior-work baselines.

On **FPGAs**, our framework enables rapid exploration of latency, DSP usage, and clock-frequency tradeoffs.

Table 3: Locus vs. Prior ASICs

Curve	Design	Freq.	Latency (ns)	Area (mm²)	Power (W)
BN254	Baseline HLS	1 GHz	63	5.10	4.72
	Locus Fastest	1 GHz	27 (2.33×)	2.21 (2.31×)	5.13
	Locus Smallest	1 GHz	35 (1.80×)	1.76 (2.90×)	3.64
BLS12-381	Baseline HLS	1 GHz	82	11.94	8.26
	Locus Fastest	1 GHz	31 (2.65×)	4.65 (2.57×)	6.83
	Locus Smallest	1 GHz	49 (1.67×)	3.88 (3.08×)	7.11
BLS12-377	Baseline HLS	1 GHz	82	11.70	8.12
	Locus Fastest	1 GHz	31 (2.65×)	3.88 (3.01×)	5.87
	Locus Smallest	1 GHz	44 (1.86×)	3.54 (3.30×)	7.15
MNT4753	Baseline HLS	667 MHz	169.5	45.89	21.76
	Locus Fastest	1 GHz	51 (3.32×)	24.00 (1.91×)	44.16
	Locus Smallest	1 GHz	68 (2.49×)	14.43 (3.18×)	23.04

Table 4: Locus vs. Prior FPGA - BLS12-377*

Design	Model	LUTs	Regs	DSPs	Cycles	Fmax (MHz)	Latency (μs)
Locus Fastest	VU9P	313,438	262,880	4,361	50	291	0.172
Locus Min-DSP	VU9P	498,716	369,423	2,436	52	259	0.201
Locus Balanced	VU9P	416,597	289,739	3,402	53	277	0.191
Locus High-Fmax	VU9P	505,704	437,901	4,032	75	408	0.184
Locus Fastest	VH1782	344,658	296,964	4,592	51	308	0.166
Locus Min-DSP	VH1782	569,452	417,245	1,512	69	241	0.286
Locus Balanced	VH1782	355,861	289,557	2,898	65	279	0.233
CycloneMSM [1]	VU9P	310,717	337,944	2,268	96	250	0.384
HardcamlMSM [29]	VU9P	–	–	–	238	278	0.856
BSTMSM [40]	U250	–	–	–	260	300	0.867

Future Work

We are developing **protocol-level kernels**, such as **Sumcheck** and **MSM**, and **full protocol accelerators**, building on the existing hardware library and framework. We are also developing **formal verification** capabilities to strengthen correctness guarantees as the IP library expands.

References

- [1] Gaurav Kuwar, Alhad Daftardar, Jianqiao Mo, Siddharth Garg, and Brandon Reagen. "Locus: A Framework for Exploring and Optimizing Point Addition Hardware for Zero-Knowledge Proofs." IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026.
- [2] Gaurav Kuwar, Alhad Daftardar, Jianqiao Mo, and Brandon Reagen. "Tessera: An Open-Source Hardware Framework for Cryptographic Kernels." Fifth Workshop on Open-Source Computer Architecture Research (OSCAR), 2026.

Acknowledgements

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Website



Github Repo

